Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

WINTER SEM-2025-2026

COURSE PROJECT

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Reg no: 23MIS0616

Programme : M.tech(SE)

Course Title : Agile Development Process and DevOps

Course Code : ISWE40L

Faculty Name : Bala Ganesh N

SMART PARKING SYSTEM

Abstract :

The Smart Parking System is developed to tackle the increasing challenges of parking space management in urban areas by offering an efficient, automated, and user-friendly solution. With the rapid growth in the number of vehicles, traditional parking methods often result in congestion, time delays, and user inconvenience. This system aims to optimize the utilization of available parking spaces while simplifying the overall parking allocation and monitoring process.

The Smart Parking System efficiently manages parking spaces and helps reduce traffic congestion in busy environments. It enables users to check real-time availability of parking slots, reserve spaces in advance, and seamlessly manage vehicle entry and exit operations. The system incorporates CRUD operations to ensure accurate and reliable data handling, along with JUnit testing to maintain system robustness. It is deployed as a microservice architecture supported by a CI/CD pipeline, using Docker for containerization and Kubernetes for scalability. Overall, the system enhances parking efficiency, reduces operational complexity, and improves user convenience.

Problem statement :

Increased vehicle usage has made parking management inefficient, causing congestion, time loss, and frustration due to lack of real-time availability and proper slot allocation. The problem is to develop a Smart Parking System that enables efficient tracking, reservation, and management of parking spaces with accurate and reliable data handling.

Objectives :

- To provide real-time information on parking slot availability
- To enable users to reserve parking spaces easily
- To manage vehicle entry and exit efficiently
- To ensure accurate storage and retrieval of parking data (CRUD operations)
- To implement testing using JUnit for system reliability
- To deploy the system using Docker and CI/CD pipeline for scalability and automation

Tools and Technologies used :

- **Programming Language:** Java
- **IDE:** Visual Studio Code
- **Build Tool:** Apache Maven
- **Testing:** JUnit
- **Version Control:** Git & GitHub
- **CI/CD:** GitHub Actions / Jenkins
- **Containerization:** Docker
- **Orchestration:** Kubernetes
- **Web Server:** Apache Tomcat
- **Data Storage:** In-memory / File system (no database used)

Project Structure :

Task 1

Source Code Implementation :

Main code :

ParkingSlot.java 1

src > java > com > contact > model > ParkingSlot.java > ParkingS

```java
package com.parking.model;

public class ParkingSlot {
    private int slotId;
    private boolean occupied;
    private String vehicleNumber;

    public int getSlotId() { return slotId; }
    public void setSlotId(int slotId) { this.slo

    public boolean isOccupied() { return occupie
```

DAO CODE

ParkingDAO.java 1, M

src > java > com > contact > model > dao > ParkingDAO.java

```java
package com.parking.dao;
// test pipeline-
import com.parking.model.ParkingSlot;
import java.util.*;

public class ParkingDAO {

    private static List<ParkingSlot> slots

    static {
        for(int i=1;i<=5;i++){
            ParkingSlot s = new ParkingSlot
            s.setSlotId(i);
            s.setOccupied(false);
            slots.add(s);
        }
    }

    // Park vehicle
    public String parkVehicle(String vehicl
        for(ParkingSlot s : slots){
            if(!s.isOccupied()){
                s.setOccupied(true);
                s.setVehicleNumber(vehicleN
                return "Parked at slot " +
            }
        }
        return "Parking Full";
    }

    // Remove vehicle
    public String removeVehicle(int slotId)
```

PARKING SLOT

ParkingSlot.java 1

src > java > com > contact > model > ParkingSlot.java > ParkingSlot

```java
package com.parking.model;

public class ParkingSlot {
    private int slotId;
    private boolean occupied;
    private String vehicleNumber;

    public int getSlotId() { return slotId; }
    public void setSlotId(int slotId) { this.slotId = s

    public boolean isOccupied() { return occupied; }
    public void setOccupied(boolean occupied) { this.oc
```

SERVLET CODE

ParkingServlet.java 1, U

src > java > com > contact > model > dao > servlet > ParkingServlet.java > Park

```java
package com.parking;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.*;

@WebServlet("/parking")
public class ParkingServlet extends HttpServlet {

    private static List<String> slots = new ArrayList<>();

    @Override
    protected void doGet(HttpServletRequest req, HttpServle
            throws ServletException, IOException {

        resp.setContentType(type: "text/html");
        PrintWriter out = resp.getWriter();

        out.println(x: "<h2>Available Slots:</h2>");
        for (int i = 1; i <= 10; i++) {
            if (!slots.contains("Slot " + i)) {
                out.println("Slot " + i + " is FREE <br>");
            } else {
                out.println("Slot " + i + " is OCCUPIED <br
            }
        }
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServl
            throws ServletException, IOException {

        String slot = req.getParameter(name: "slot");

        if (slot != null && !slots.contains(slot)) {
```

Task 2

JUNIT TESTING

parkingtest.java 1

src > java > com > contact > parkingtest.java > ParkingTest

```java
package com.parking;

import com.parking.dao.ParkingDAO;
import org.junit.Test;
import static org.junit.Assert.*;

public class ParkingTest {

    @Test
    public void testParking() {
        ParkingDAO dao = new ParkingDAO();
        String result = dao.parkVehicle("AP01AB1234");

        assertTrue(result.contains(s: "Parked"));
    }
}
```

```
PS C:\Users\hp\ contact-app> mvn clean package
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ contact-app ---
[INFO] Deleting C:\Users\hp\ contact-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ cont
[INFO] skip non existing resourceDirectory C:\Users\hp\ contact
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ contact-
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources
[INFO] skip non existing resourceDirectory C:\Users\hp\ contact
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ contact-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ contact-app ---
[INFO] Packaging webapp
[INFO] Assembling webapp [contact-app] in [C:\Users\hp\ contact
```

The junit test cases were executed successfully using maven.

The output shows that all test cases passes with zero failures and errors.

This confirms that the logic is working correctly.

Task 3

CI/CD Pipeline

ci.yml

.github > workflows > ci.yml

```yaml
name: Smart Parking CI

on:
  push:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
    - uses: actions/checkout@v4

    - name: Setup Java
      uses: actions/setup-java@v4
      with:
        distribution: temurin
        java-version: '17'

    - name: Build
      run: mvn clean package

    - name: Test
      run: mvn test
```

ci.yml

.github > workflows > ci.yml

```yaml
name: Smart Parking CI

on:
  push:
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  build:
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    - name: Setup Java
      uses: actions/setup-java@v4
      with:
        distribution: temurin
        java-version: '17'

    - name: Build
      run: mvn clean package

    - name: Test
      run: mvn test
```
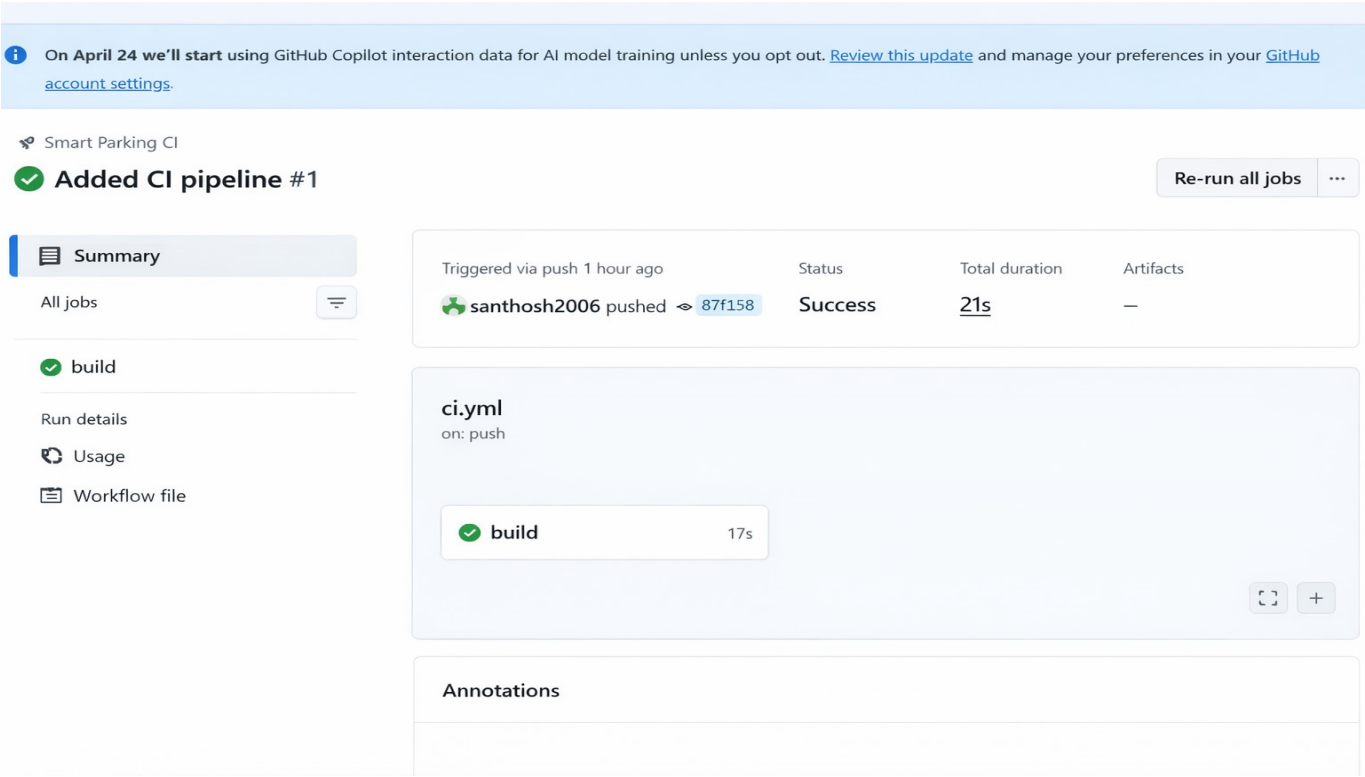



CI/CD pipline is implemented using GitHub Actions.

The pipeline automatically builds the project and executes test cases whenever code is pushed to GitHub.

Task 4

Docker implementation

```
PS C:\Users\hp\ contact-app> docker build -t parking-app:latest .
[+] Building 4.7s (8/8) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from dockerfile                                                0.1s
 => => transferring dockerfile: 110B                                                                0.1s
 => [internal] load metadata for docker.io/library/tomcat:9.0                                       3.2s
 => [auth] library/tomcat:pull token for registry-1.docker.io                                       0.0s
 => [internal] load .dockerignore                                                                   0.0s
 => => transferring context: 2B                                                                     0.0s
 => [internal] load build context                                                                   0.0s
 => => transferring context: 70B                                                                    0.0s
 => [1/2] FROM docker.io/library/tomcat:9.0@sha256:6257fbec6ef16433a50233215a4fcf50564bf5b8fb1a92f8656460f95a98d5e4   0.1s
 => => resolve docker.io/library/tomcat:9.0@sha256:6257fbec6ef16433a50233215a4fcf50564bf5b8fb1a92f8656460f95a98d5e4   0.1s
 => CACHED [2/2] COPY target/contact-app.war /usr/local/tomcat/webapps/                             0.0s
 => exporting to image                                                                              0.5s
 => => exporting layers                                                                             0.0s
 => => exporting manifest sha256:3afaeebae4868ddb453e3a0407fcd1fca47539b1861432e1d45636a240108bb0   0.0s
 => => exporting config sha256:09e71b417d2218e0fb3882f66b70ec09e91b0c85486b595190574341ca5b9f98     0.0s
 => => exporting attestation manifest sha256:010ce760d10823ad2b02f8f9511b852904a34f22eb8caae450eb761e92dfe848   0.1s
 => => exporting manifest list sha256:e7ced332e1a1c600acb35ac4987fa64ade7e8a1565b36054027f5c3663dc5e9d   0.0s
 => => naming to docker.io/library/parking-app:latest                                               0.0s
 => => unpacking to docker.io/library/parking-app:latest                                            0.0s
PS C:\Users\hp\ contact-app>
```

```
PS C:\Users\hp\ contact-app> docker images
                                                                                                  i Info →   U   In Use
IMAGE                                                                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
balaganesh81/java-test-app:latest@sha256:e7bf2df109f934ecd4361622af1ddb7fee2341d2f95783242e24ea991a3a7166
                                                                                        e7bf2df109f9       170MB          170MB
chandanatr/parking-app:latest                                                           775b4a2254af       585MB          154MB
contact-app:latest                                                                      fb1f800abe4c       585MB          154MB    U
docker/desktop-cloud-provider-kind:v0.5.0                                               4ad59ce20658       595MB          162MB    U
docker/desktop-containerd-registry-mirror:v0.0.3                                        179a9ea3ae47      50.3MB         16.1MB    U
envoyproxy/envoy:v1.36.4                                                                08acc53db6f4       248MB         62.1MB
gcr.io/k8s-minikube/kicbase:v0.0.50                                                     b97074569ae9      1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93
                                                                                        eb4fec00e8ad      1.94GB          545MB    U
kindest/node:v1.34.3                                                                    08497ee19eac      1.35GB          395MB    U
parking-app:latest                                                                      e7ced332e1a1       585MB          154MB
PS C:\Users\hp\ contact-app>
```

Book Parking Slot

Enter Slot Number (1-10): 1

Book Slot

Check Parking Status

View All Slots

Free a Slot

Enter Slot Number: 12

Free Slot

```
C:\Users\hp\.jenkins\workspace\parking-pipeline>docker run -d -p 7075:8080 --name parking-container parking-app
8a1d5f31a5322095865b00a0caf8ad15e1a84fe821879f6772e9cd7ae0b20cb8
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Timings

		Primary task	Including subtasks
In queue	Waiting	52 ms	56 ms
	Blocked	0 ms	0 ms
	Buildable	0 ms	14 ms
	Total	0.11 sec	0.13 sec
Building		49 sec	43 sec
Scheduled to completion		49 sec	
Number of subtasks		1	
Average executor utilization		0.9	

Containers Slew Feedback

Container CPU usage
0.23% / 800% (8 GPUS available)

Container memory usage
2836.7MiB / 3.64GiB

Show charts

Search

Only show running containers

Name	Container ID	Image	Port(s)	CPU(%)	Last started
firewor_islande	943f5795C376	contact-app	7070:8080	0%	
zealous_bejevp	5efcd9dbe94f	contact-app	7070:8080	0%	
peaceful_danko	a3d879561791	contact-app	7070:8080	0%	
stupef.linkman	9800f648663	contact-app	7070:8080	0%	
relaxed-imbram	86cc3d1a32f1	contact-app	7070:8080	0%	
distracting-gender	9f38fc6a8af8	contact-app	7070:8080	0%	
cheerful_whate	963854d5863	contact-app	7070:8080	0%	
friendly_talcnover	000700029656	parking-app	7070:8080 c2	1.44%	21 minutes ago
parking-container	0b5fa132fae0	parking-app		1.4%	

Showing 14 kermes

Jenkins / parking pipeline #7 Stages

#77

Re-run

Started by Santhosh S Started 16 min 33 sec ago e1cd266 Took 48 sec

Graphs

Start — Checkout SCM — Build — Test — Docker Build — Run Container — End

Search

Run Container 0 Started 16 min ago 48s

Docker SCM
100%
55%
RSM
Docker Build
Run Container

Windows Batch Script `docker stop parking container | > NUL 2&&1 | > docker` iscr

```
1
2 C:\Users\xyL_jenkins\workspace\parking-pipeline>docker stop parking-container
3 Error response from daemon: No such container: parking-container
4 no container
5
5 C:\Users\xyL_jenkins\workspace\parking-pipeline>docker rmm parking-container
7 Error response from daemon: No such container:
8 no container
1
```

Kubernets deployement

service.yaml U

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: parking-service
spec:
  type: NodePort
  selector:
    app: parking-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30007
```

deployement.yaml M

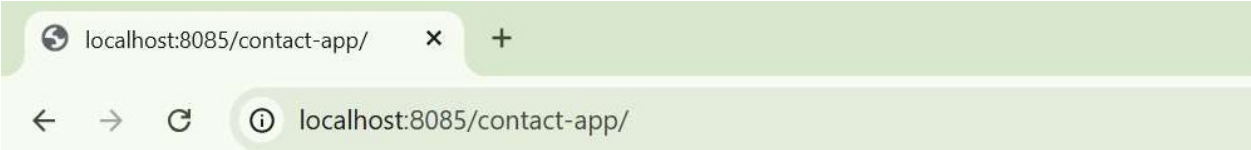
deployement.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: parking-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: parking
  template:
    metadata:
      labels:
        app: parking
    spec:
      containers:
        - name: parking-app
          image: chandanatr/parking-app:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 8080
```

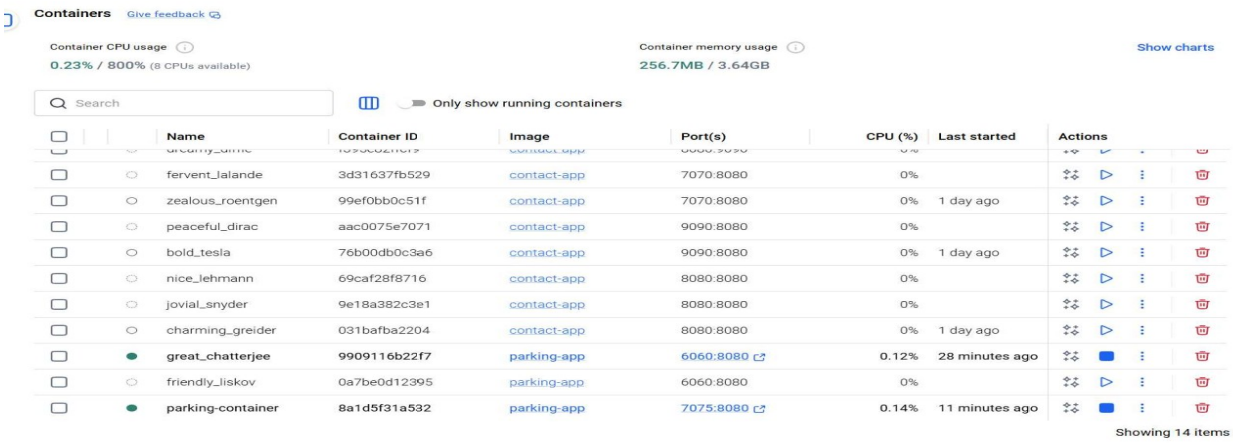
```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\hp\ contact-app> kubectl get services
NAME              TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
kubernetes        ClusterIP   10.96.0.1     <none>        443/TCP        64m
parking-service   NodePort    10.96.54.70   <none>        80:30007/TCP   62m
PS C:\Users\hp\ contact-app> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
parking-deployment-79758599f8-qmstv   1/1     Running   0          31m
PS C:\Users\hp\ contact-app> kubectl get deployments
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
parking-deployment   1/1     1            1           32m
PS C:\Users\hp\ contact-app>
```

Output :

Application running



Docker ouput



Name	Container ID	Image	Port(s)	CPU (%)	Last started
fervent_lalande	3d31637fb529	contact-app	7070:8080	0%	
zealous_roentgen	99ef0bb0c51f	contact-app	7070:8080	0%	1 day ago
peaceful_dirac	aac0075e7071	contact-app	9090:8080	0%	
bold_tesla	76b00db0c3a6	contact-app	9090:8080	0%	1 day ago
nice_lehmann	69caf28f8716	contact-app	8080:8080	0%	
jovial_snyder	9e18a382c3e1	contact-app	8080:8080	0%	
charming_greider	031bafba2204	contact-app	8080:8080	0%	1 day ago
great_chatterjee	9909116b22f7	parking-app	6060:8080	0.12%	28 minutes ago
friendly_liskov	0a7be0d12395	parking-app	6060:8080	0%	
parking-container	8a1d5f31a532	parking-app	7075:8080	0.14%	11 minutes ago

```
PS C:\Users\hp\ contact-app> docker images
                                                                                                  i Info →   U  In Use
IMAGE                                                                                   ID            DISK USAGE   CONTENT SIZE   EXTRA
balaganesh81/java-test-app:latest@sha256:e7bf2df109f934ecd4361622af1ddb7fee2341d2f95783242e24ea991a3a7166
                                                                                        e7bf2df109f9       170MB          170MB
chandanatr/parking-app:latest                                                           775b4a2254af       585MB          154MB
contact-app:latest                                                                      fb1f800abe4c       585MB          154MB    U
docker/desktop-cloud-provider-kind:v0.5.0                                               4ad59ce20658       595MB          162MB    U
docker/desktop-containerd-registry-mirror:v0.0.3                                        179a9ea3ae47      50.3MB         16.1MB    U
envoyproxy/envoy:v1.36.4                                                                08acc53db6f4       248MB         62.1MB
gcr.io/k8s-minikube/kicbase:v0.0.50                                                     b97074569ae9      1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93
                                                                                        eb4fec00e8ad      1.94GB          545MB    U
kindest/node:v1.34.3                                                                    08497ee19eac      1.35GB          395MB    U
parking-app:latest                                                                      e7ced332e1a1       585MB          154MB
PS C:\Users\hp\ contact-app>
```

Docker container runs successfully

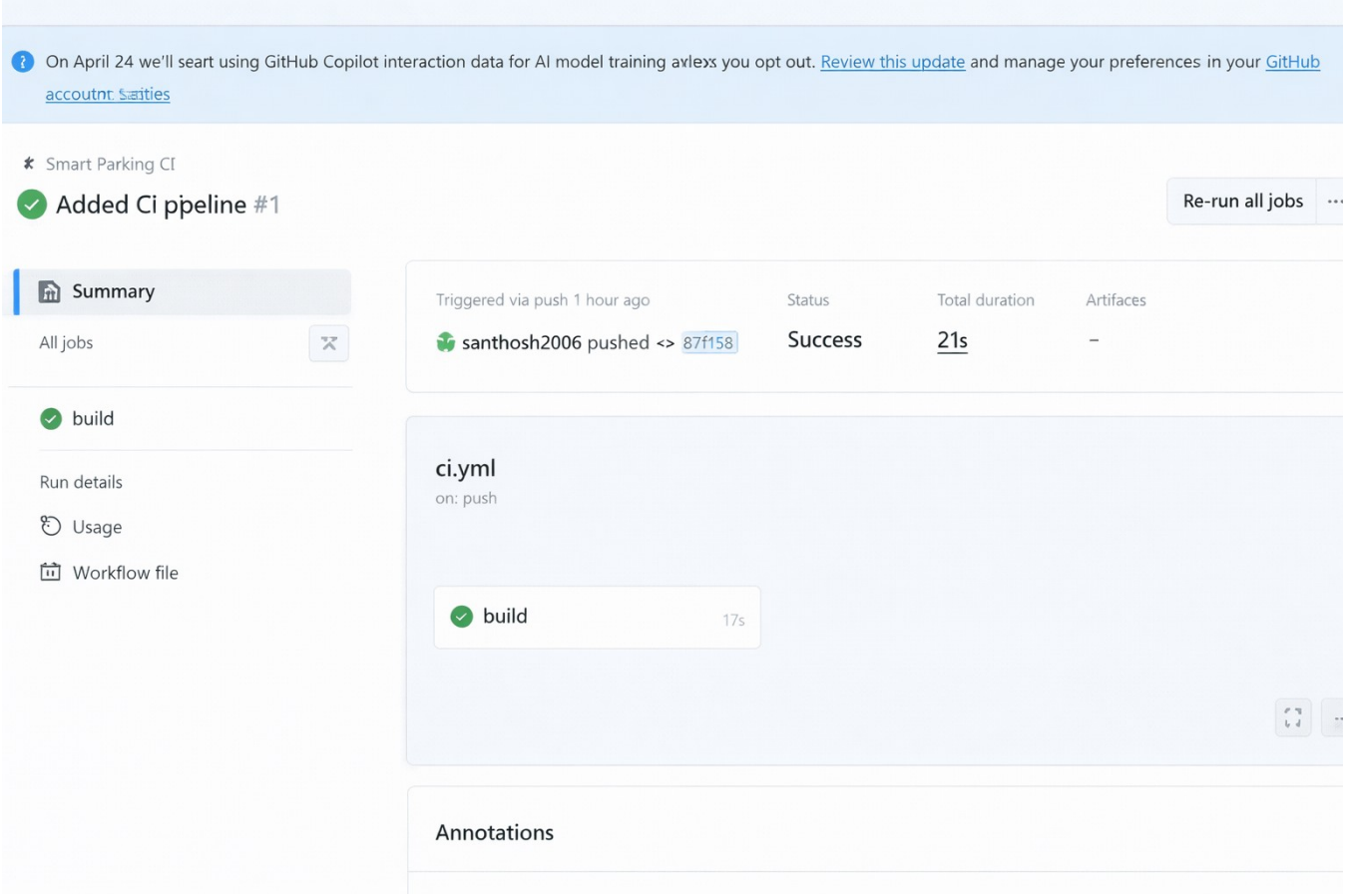
Kubernets output

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\hp\ contact-app> kubectl get services
NAME             TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
kubernetes       ClusterIP   10.96.0.1     <none>        443/TCP        64m
parking-service  NodePort    10.96.54.70   <none>        80:30007/TCP   62m
PS C:\Users\hp\ contact-app> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
parking-deployment-79758599f8-qmstv   1/1     Running   0          31m
PS C:\Users\hp\ contact-app> kubectl get deployments
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
parking-deployment   1/1     1            1           32m
PS C:\Users\hp\ contact-app>
```

Junit output

```
PS C:\Users\hp\ contact-app> mvn clean package
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ contact-app ---
[INFO] Deleting C:\Users\hp\ contact-app\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ cont
[INFO] skip non existing resourceDirectory C:\Users\hp\ contact
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ contact-
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources
[INFO] skip non existing resourceDirectory C:\Users\hp\ contact
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ contact-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ contact-app ---
[INFO] Packaging webapp
[INFO] Assembling webapp [contact-app] in [C:\Users\hp\ contact
```

CI/CD output



Conclusion :

The smart parking system successfully developed and deployed. It demonstrates an automated and efficient approach to managing parking slot allocation using web-based technology.This project was developed using Java Servlets for backend logic and deployed as a web application using Apache Tomcat. To achieve a modern DevOps workflow, the application was containerized using Docker, ensuring portability and consistency across environments. Further, Kubernetes was used to deploy and manage the application in a cluster, enabling scalability and reliable service availability. Jenkins was integrated to automate the build and deployment process, implementing a complete CI/CD pipeline.